Appendix C: IEEE 830 Template

Software Requirements Specification

for

Tamycs Shield

Version v1

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Revision History

Name	Date	Reason for Changes	Version
YAYA ABDEL SATAR	4 MARCH	Initial version V1 Initial Requirements Analysis	V1

YAYA ABDEL SATAR	11 MARCH	UseCase and description	V2

1. Problem Statement

Current Situation and Problem:

Academic institutions increasingly rely on digital platforms for conducting examinations. However, standard assessment tools face two critical challenges. First, they are highly vulnerable to network instability; a simple Wi-Fi disconnection can cause a student to lose all their unsaved progress, leading to unfair grading and administrative overhead. Second, they often lack robust anti-cheating mechanisms and data security, making them susceptible to basic exploits (such as SQL injections) or unauthorized access to local test files.

Proposed Solution (Tamycs Shield):

To address these issues, we propose Tamycs Shield, a secure, 3-tier examination system. The system utilizes a modern technology stack to ensure both resilience and security. By implementing automated local caching encrypted with AES-256-GCM, the system guarantees that no student data is lost during network drops. Furthermore, the application will operate in a locked-down environment to prevent unauthorized access to external resources. This project aims to deliver a tamper-proof, highly reliable digital examination environment for universities.

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2. Overall Description

Product Perspective Tamycs Shield is a self-contained, 3-tier web application designed to replace legacy monolithic assessment tools. It consists of:

Presentation Layer (Client): A dynamic interface built with React for students.

Application Logic Layer (Server): A Node.js backend handling API routing and security logic.

- Data Layer (Database): A PostgreSQL relational database.

User Classes

Administrators (Teachers): Users who create exams, manage question databases, and review student grades.

- Students: End-users who interact exclusively with the secure client interface to take exams.

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3. Functional Requirements

Note: All requirements are assigned a unique ID for traceability and use the keyword "shall" to indicate an engineering obligation.

- FR-01: The system shall allow Administrators to create an account using a valid university email address and a password.
- FR-02: The Tamycs Server shall sanitize all text inputs from Administrators before storing them in the PostgreSQL database to prevent SQL injection attacks.
- FR-03: Upon the initiation of an exam, the Tamycs Client shall randomly shuffle the order of the questions to generate a unique test sequence for each student.
- FR-04: The Tamycs Client shall automatically save the student's current answers to an encrypted local backup file every 60 seconds.
- FR-05: If the network connection is lost, the Tamycs Client shall attempt to reconnect to the server every 30 seconds without interrupting the student's exam timer.
- FR-06: The system shall calculate and display the final grade to the Administrator within 5 seconds after a student submits their exam.

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4. Non-Functional Requirements

Note: These requirements define how well the system performs, focusing on security, performance, and reliability.

- NFR-01 (Security): All user and administrator passwords shall be hashed using the bcrypt algorithm before being stored in the database.
- NFR-02 (Performance): The system shall respond to client API requests (such as submitting an answer) in less than 2 seconds.
- NFR-03 (Reliability): The Tamycs Client application shall maintain a crash-free session rate of 99.5% during a standard 2-hour examination period.
- NFR-04 (Security): Data transmission between the React frontend and Node.js backend shall be encrypted using standard HTTPS/TLS protocols.

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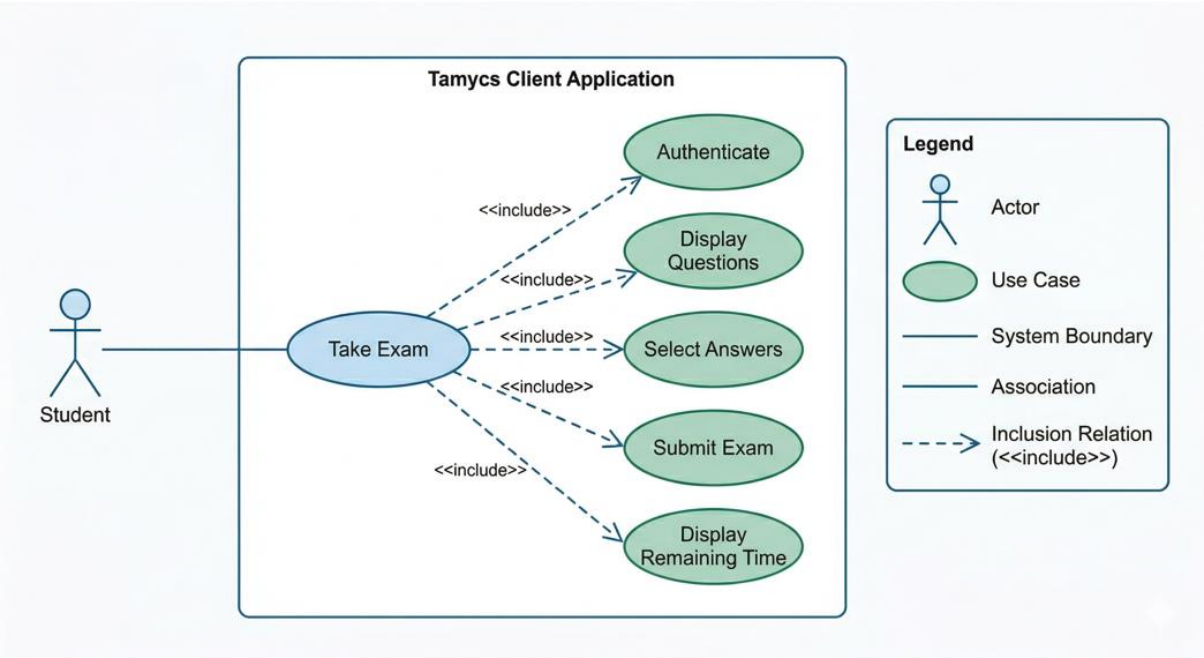
5. External Interfaces

5.1 User Interfaces Overview The system features a Graphical User Interface (GUI). The student interface (Tamycs Client) shall operate in a full-screen, distraction-free mode displaying only the current question, multiple-choice answers, and a countdown timer.

5.2 Software Interfaces The React client shall communicate with the Node.js server exclusively via RESTful API endpoints using JSON formatting for all data payloads.

UseCase



6 . System: Tamycs Client Application

This diagram illustrates the functional interactions between the student and the secure examination client interface.

A. Actors Identification

- **Primary Actor: Student** – The end-user who interacts with the React-based client interface to complete an assessment. The student is linked to the core functionality via an **Association** line.

B. Use Case Descriptions

The diagram breaks down the examination process into several specialized modules:

- **Take Exam**: The base use case initiated by the student to start the testing session.
- **Authenticate**: A mandatory security step requiring a valid university email and password before access is granted.
- **Display Questions**: The interface presents the test items, which are randomized to ensure a unique sequence for every student.

- **Select Answers**: The functional requirement allowing the student to choose multiple-choice options.
- **Submit Exam**: The final action where the student transmits their completed test to the Node.js server for grading.
- **Display Remaining Time**: A constant visual countdown timer to assist with student time management.

C. Logical Relations (UML)

The diagram relies exclusively on **Inclusion** relationships to define the system workflow:

- **`<<include>>`**: All dashed arrows point from the **Take Exam** case toward the other specific use cases.
- **Engineering Obligation**: This indicates that the "Take Exam" process cannot be completed without executing all the included sub-steps. In UML terms, these behaviors are required parts of the base use case and are not optional.